

Moving from Enterprise Value to Value per Share

When you have completed the valuation of core operations, as described in Chapter 10, you are ready to estimate enterprise value, equity value, and value per share. Enterprise value represents the value of the entire company, while equity value represents the portion owned by shareholders.

To determine enterprise value, add nonoperating assets to the value of core operations. The most common nonoperating assets are excess cash, investments in nonconsolidated companies, and tax loss carryforwards.¹ To estimate equity value, subtract all nonequity claims from enterprise value. Nonequity claims include short-term and long-term debt, debt equivalents like unfunded pension liabilities, and hybrid securities like convertible securities and employee stock options. Finally, to estimate the intrinsic value per share, divide the resulting equity value by the most recent number of shares outstanding.

While nonoperating assets and nonequity claims may feel like an afterthought, this is not the case. Many sophisticated investors have discovered substantial value hidden in nonoperating assets, especially in privately held conglomerates. In contrast, other investors have been burned by not accurately identifying and valuing all nonequity claims against enterprise value, as happened in the well-publicized case of Enron. It is critical to know who has a claim on cash flow before equity holders do.

This chapter lays out the process for converting core operating value into enterprise value and subsequently into equity value. The chapter goes

¹ Throughout the book, we define enterprise value as the value of core operations plus nonoperating assets. Many bankers define enterprise value as debt plus equity minus cash. For a company whose only nonoperating asset is excess cash and owes only traditional debt, this definition is equivalent to our definition of the value of core operations. This simple definition of enterprise value, however, fails to account for other nonoperating assets and debt equivalents, which can lead to errors in valuation.

step-by-step through the process of identifying and valuing the most common nonoperating assets, debt and debt equivalents, hybrid securities, and noncontrolling interests, ending with the final step in valuation—estimating the intrinsic value per share.²

THE VALUATION BUILDUP PROCESS

The valuation buildup begins with a company's core operating value, based on discounted cash flow (DCF)—the top line of the example shown in Exhibit 16.1. This amount plus nonoperating assets equals enterprise value. The equity value—the bottom line in the exhibit—is the value that remains after subtracting from the enterprise value all the nonequity claims, which include interest-bearing debt, debt equivalents, and hybrid claims. We use the term *nonequity claim* because there are many financial claims against a company's cash flows other than traditional fixed-coupon debt and shareholders' equity.

EXHIBIT 16.1 **Sample Comprehensive Valuation Buildup**

\$ million		
DCF value of operations	5,000	
Excess cash and marketable securities	50	Nonoperating assets
Excess real estate	5	
Investments in nonconsolidated companies	270	
Financial subsidiary	300	
Tax loss carryforwards	10	
Discontinued operations	30	
Enterprise value	5,665	
Claims against enterprise value		
Bank loans	(250)	Interest-bearing debt
Bonds	(550)	
Long-term operating provisions	(50)	Debt equivalents
Nonoperating provisions	(75)	
Operating leases	(250)	
Unfunded retirement obligations	(200)	
Contingent liabilities	(40)	
Convertible debt	(200)	Hybrid claims and noncontrolling interests
Preferred stock	(100)	
Employee stock options	(50)	
Noncontrolling interests	(150)	
Equity value	3,750	

² Estimating the value per share completes the technical aspect of the valuation, yet the job is not complete. It is then time to revisit the valuation with a comprehensive look at its implications. We examine this process in Chapter 17.

In general, a nonoperating asset is any asset that you have not incorporated as part of free cash flow. Common nonoperating assets are excess cash, one-time receivables, investments in nonconsolidated companies (also known as equity investments and by other names), excess pension assets, discontinued operations, and financial subsidiaries. Take extra care not to classify an asset required for ongoing operations as nonoperating. For instance, some analysts who follow retailers add the value of real estate to the value of core operations. Since the real estate is required to conduct business, its benefits are already embedded in the value of operations. The value of real estate can only be added to core operations if the company is charged a market-based rent in free cash flow. Otherwise, including the value of real estate will lead to an overestimate of value.

Nonequity claims are financial claims against enterprise value whose expenses are not included in EBITA and consequently are excluded from free cash flow. Traditional debt contracts like bank debt and corporate bonds are the most common nonequity claims. Other debt-like claims, known as debt equivalents, include the present value of operating leases, unfunded pension and other retirement liabilities, and environmental remediation liabilities, among others. Because these claims do not scale with revenue or can affect the cost of capital, they are best valued separately from free cash flow.

Nonequity claims also include hybrid securities, such as preferred stock, convertible securities, and employee options, which have characteristics of both debt and equity. Such hybrids require special care: their valuations are highly dependent on enterprise value, so you should value them using option-pricing models rather than book value.³ Finally, if other shareholders have noncontrolling interests against certain consolidated subsidiaries, deduct the value of the noncontrolling interests to determine equity value. Like hybrid securities, noncontrolling interests will correlate with enterprise value, so extra care is required.

VALUING NONOPERATING ASSETS

Although not included in free cash flow, nonoperating assets still represent value to the shareholder. Thus, to arrive at enterprise value, you must estimate the market value of each nonoperating asset separately and add the resulting value to the DCF value of operations. If necessary, adjust for circumstances that could affect shareholders' ability to capture the full value of these assets. For example, if the company has announced it will sell off a nonoperating asset in the near term, deduct the estimated capital gains taxes (if any) on the asset from its market value. If ownership of the asset is shared with another company, include only your company's portion of the value.

³For investment-grade companies, the value of debt is driven mostly by interest rates. In this case, there will be little interdependence between enterprise value and debt. For distressed companies, default risk also drives the value of debt. In this case, interdependence will be high and must be modeled. We discuss highly levered companies later in the chapter.

This section identifies the most common nonoperating assets and describes how to handle each of them in the valuation.

Excess Cash and Marketable Securities

As discussed in Chapter 11, companies often hold more cash and marketable securities than they need to run the business. Companies hold excess cash for a number of reasons, parking it in short-term securities until they can invest it or return it to shareholders. Prior to the change in American tax laws in 2018, American companies held significant amounts of excess cash when they had substantial earnings outside the United States. They were reluctant to repatriate cash because they were required to pay any difference in taxes upon repatriation. With a drop in the corporate tax rate from 35 percent to 21 percent, many companies have committed to repatriating cash. How they deploy this cash will unfold over time, but it will probably consist of new investment, increased dividends, and significant share repurchases.⁴

You should make an estimate of how much the business needs for operations. The remaining cash and marketable securities are treated as nonoperating. As a rule of thumb, we often assume that a company requires about 2 percent of revenues in cash to operate the business. The remaining cash and marketable securities are considered excess.

Cash and marketable securities are reported on a company's balance sheet at fair market value. You can use these assets' book value in your valuation, unless you have reason to believe they have significantly changed in value since the reporting date (as in the limited case of volatile equity holdings).

Investments in Nonconsolidated Companies

Companies often invest in other companies without taking control, and hence they do not consolidate the investment's financial statements into their own. Investments in nonconsolidated companies can be found on the balance sheet under many names. For instance, Philips reports its investments in nonconsolidated companies as investments in associates, Intel reports them as equity investments, and PPG Industries reports them as investment in equity affiliates.

Because the parent company does not have control over these subsidiaries, their financials are not consolidated, so these investments must be valued separately from operations. Under U.S. Generally Accepted Accounting Principles (GAAP) and International Financial Reporting Standards (IFRS),

⁴ For examples of repatriation and redeployment, see A. Balakrishnan, "Apple Announces Plans to Repatriate Billions in Overseas Cash, Says It Will Contribute \$350 Billion to the US Economy over the Next 5 Years," CNBC, January 17, 2018, www.cnbc.com. For more on share buybacks, see K. Rooney, "Share Buybacks Soar to Record \$806 Billion—Bigger Than a Facebook or Exxon Mobil," CNBC, March 25, 2019, www.cnbc.com.

there are two ways in which nonconsolidated subsidiaries can appear in the parent company's accounts:

1. For equity stakes in which the parent company exerts "significant influence" but lacks control (often between 20 and 50 percent ownership), the equity holding in the subsidiary is reported on the parent's balance sheet at the investment's historical cost plus any reinvested income. The parent company's portion of the subsidiary's profits is shown on the parent's income statement as other income, unless specifically disclosed. Accountants refer to this as the equity method.
2. For equity stakes below 20 percent, the parent company is often assumed to have no influence. The equity holdings are shown at historical cost on the parent's balance sheet. The parent's portion of the subsidiary's *dividends* is included in other income on the income statement.

In response to the accounting and financial scandals of the early 2000s, global accounting moved away from absolute thresholds to consolidation methods that rely on the parent's influence around key activities and exposure to gains and losses. Implementation has been complex, and companies can report under multiple standards. Always investigate the notes to determine which investments contribute to EBITA and which do not.

Investments in Publicly Traded Companies If an investment in another company is publicly listed, use the market value to determine the value of the parent company's equity stake. Verify that the market value is indeed a good indicator of intrinsic value. In some cases, these listed subsidiaries have very limited free float and/or very low liquidity, so the share price may not properly reflect current information.

Exhibit 16.2 presents the equity investments held by Coca-Cola. Since Coca-Cola does not control these companies, their revenue, income, and assets are not consolidated on Coca-Cola's financial statements. Therefore, each investment must be valued separately and added to Coca-Cola's value of operations to determine enterprise value.

In the management discussion and analysis section of its 2018 annual report, Coca-Cola reports both book value and fair value for its equity investments.⁵ Therefore, if you are valuing Coca-Cola near its fiscal-year close, the valuation from the annual report will suffice. As the year progresses, however, these data become stale, and each investment must be revalued. For example, consider Coca-Cola Amatil, Coca-Cola's bottler in Australia. To value this

⁵ Companies will disclose how fair value is determined for each security, using a system of levels. Level 1 inputs are quoted prices of identical securities in liquid markets. Level 2 inputs are quoted prices of identical securities in illiquid markets or similar securities in liquid markets. Level 3 inputs are not observed and are estimated using financial models.

EXHIBIT 16.2 **Coca-Cola Company: Publicly Traded Equity Investments, December 2018**

\$ million

	Book value	Fair value	Valuation of Coca-Cola Amatil Limited (ASX: CCL)	
Monster Beverage Corporation	3,573	5,026	Share price, AU \$	8.19
Coca-Cola European Partners plc	3,551	4,033	× Shares outstanding, million	724
Coca-Cola FEMSA, S.A.B. de C.V.	1,714	3,401	= Market capitalization, AU \$ million	5,930
Coca-Cola HBC AG	1,260	2,681		
Coca-Cola Amatil Limited	656	1,325	← × Percent ownership	30.8%
Coca-Cola Bottlers Japan Holdings Inc.	1,142	978	= Ownership stake, AU \$ million	1,826
Embotelladora Andina S.A.	263	497		
Coca-Cola Consolidated, Inc.	138	440	× Currency conversion, US \$/AU \$	0.73
Coca-Cola İçecek A.Ş.	174	299	= Ownership stake	1,325
Total	12,471	18,680		

Source: Coca-Cola Company annual report, 2018; Coca-Cola Amatil annual report, 2018; Yahoo Finance.

equity stake, multiply the enterprise value for Coca-Cola Amatil (AU \$5,930 million) by Coca-Cola's ownership percentage (30.8 percent). The resulting ownership stake equals AU \$1,826 million. Since Coca-Cola reports in U.S. dollars, the stake must be converted into U.S. dollars at the prevailing exchange rate. Multiplying AU \$1,826 million by 0.73 equals the value of Coca-Cola's ownership of Coca-Cola Amatil (\$1,325 million).

Although this valuation was accurate as of December 31, 2018, any change in one of the inputs will require an update to the valuation. For instance, during the first quarter of 2019, Amatil's stock price rose by approximately 3 percent. This rise in value was reflected in Coca-Cola's next quarterly report but not during the interim.

Investments in Privately Held Companies If the subsidiary is not listed but you have access to its financial statements (for instance, through a public bond offering or private disclosure), perform a separate DCF valuation of the equity stake. Discount the cash flows at the appropriate cost of capital (which may be different than the parent company's weighted average cost of capital). Also, when completing the parent valuation, include only the value of the parent's equity stake and not the subsidiary's entire enterprise value or equity value.

If the parent company's accounts are the only source of financial information for the subsidiary, we suggest the following alternatives to DCF:

- *Simplified cash-flow-to-equity valuation.* This is a feasible approach when the parent has a 20 to 50 percent equity stake, because the subsidiary's net income and book equity are disclosed in the parent's accounts.⁶

⁶ The book value of the subsidiary equals the historical acquisition cost plus retained profits, which is a reasonable approximation of book equity. If goodwill is included in the book value of the subsidiary, this should be deducted.

Build forecasts for how the equity-based key value drivers (net income growth and return on equity) will develop, so you can project cash flows to equity. Discount these cash flows at the *cost of equity* for the subsidiary in question and not at the parent company's cost of capital.

- *Multiples valuation.* As a second alternative, estimate the partial stake using a price-to-earnings and/or market-to-book multiple. If the company owns 20 to 50 percent of the subsidiary, apply an appropriate multiple to reported income.
- *Tracking portfolio.* For parent equity stakes below 20 percent, you may have no information beyond the investment's original cost—that is, the book value shown in the parent's balance sheet and often only disclosed in the notes. Even applying a multiple is difficult, because neither net income nor the current book value of equity is reported. If you know when the stake was acquired (or last valued), you can approximate its current market value by applying the relative price change for a portfolio of comparable stocks over the same holding period.

You should triangulate your results as much as possible, given the lack of precision for these valuation approaches.

Loans to Other Companies

For loans to nonconsolidated subsidiaries and other companies, use the reported book value. This is a reasonable approximation of market value if the loans were given at fair market terms and if the borrower's credit risk and general interest rates have not changed significantly since issuance. If this is not the case and the investment is substantial, you should perform a separate DCF valuation of the promised interest and principal payments at the yield to maturity for corporate bonds with similar risk and maturity.

Finance Subsidiaries

To make their products more accessible, some companies operate customer financing businesses.⁷ Because financial subsidiaries differ greatly from manufacturing and services businesses, it is critical to separate revenues, expenses, and balance sheet accounts associated with the subsidiary from core operations. Failing to do so will distort return on invested capital, free cash flow, and ultimately your perspective on the company's valuation.

Once the finance subsidiary is separated from operations, use the reorganized financial statements to value the subsidiary as if it were a financial

⁷ Companies that sell expensive products typically offer financing of purchases. Significant customer financing subsidiaries exist at IBM and Textron, among others.

institution. Add this value to the value of core operations to determine enterprise value. Since the finance subsidiary's debt will already be incorporated into your valuation of the finance subsidiary, do not subtract total debt from the parent company's enterprise value to determine equity value. Subtract only general obligation debt unrelated to the finance subsidiary.

We present the valuation of a company with a finance subsidiary in Chapter 19, and we cover bank valuation in Chapter 38.

Discontinued Operations

Discontinued operations are businesses being sold or closed. The earnings from discontinued operations are explicitly shown in the income statement, and the associated net asset position is disclosed on the balance sheet. Because discontinued operations are no longer part of a company's operations, their value should not be modeled as part of free cash flow or included in the DCF value of operations. Under U.S. GAAP and IFRS, the assets and liabilities associated with the discontinued operations are written down to their fair value and disclosed as a net asset on the balance sheet, so the most recent book value is usually a reasonable approximation.⁸

Excess Real Estate

Excess real estate and other unutilized assets are assets no longer required for the company's operations. As a result, any cash flows that the assets generate are excluded from the free-cash-flow projection, and the assets are not included in the DCF value of operations. Identifying these assets in an outside-in valuation is nearly impossible unless they are specifically disclosed in the company's footnotes. For that reason, only internal valuations are likely to include their value separately as a nonoperating asset. For excess real estate, use the most recent appraisal value when it is available. Alternatively, estimate the real estate value either by using a multiple, such as value per square meter, or by discounting expected future cash flows from rentals at the appropriate cost of capital. Of course, be careful to exclude any operating real estate from these figures, because that value is implicitly included in the free-cash-flow projections and value of operations.

We do not recommend a separate valuation for unutilized operating assets unless they are expected to be sold in the near term. If the financial projections for the company reflect growth, the value of any underutilized assets should instead be captured in lower future capital expenditures.

⁸ Any upward adjustment to the current book value of assets and liabilities is limited to the cumulative historical impairments on the assets. Thus, the fair market value of discontinued operations could be higher than the net asset value disclosed in the balance sheet.

Excess Pension Assets

Surpluses in a company's pension funds show up as net pension assets on the balance sheet and typically reported at market value.⁹ (Small amounts are typically embedded within other assets.) On an after-tax basis, the pension's value depends on management's plans. If pensions are expected to be dissolved soon, subtract liquidation taxes—typically set higher than the statutory tax rate—from the market value of excess pension assets. Otherwise, subtract taxes at the statutory rate, which reflects the need for lower future contributions. For details on pension accounting and valuation, see Chapter 23.

Tax Loss Carryforwards

When a company generates a loss in a given year, it can accumulate those losses and net them against future income, thereby reducing future taxes.¹⁰ This is known as a tax loss carryforward. Since tax savings will increase future cash flows, estimate their value using discounted cash flow, and add your result to the company's value of operations.

The potential tax loss carryforward is recorded on the balance sheet as a deferred-tax asset.¹¹ Use the deferred-tax asset as a starting point to value the tax loss carryforwards. If the company is unlikely to use the tax loss carryforward, the company will record a valuation allowance against the deferred-tax asset. Both numbers can be found in the note on taxes that accompanies the company's financial statements. The company's valuation allowance will reflect its current expectations of future profitability, which may be different from your projections, so use caution in adopting the company's calculation. If you develop multiple scenarios to value operations, estimate your own allowance against the deferred-tax asset based on the probability of the asset being realized under each scenario.

Because tax savings are recorded on an undiscounted basis, apply discounted cash flow to estimate their value as of today. Ideally, you would discount tax savings at a cost of capital that perfectly matches their risk. In practice, use the weighted average cost of capital. This will appropriately counter the value of operating taxes embedded in your value of operations.

To estimate the present value, forecast the year-by-year tax savings based on your projected earnings for the company. Unless income by geography is

⁹ Under IFRS, companies can still report excess pension assets at book value. If pensions are not marked to market, search the company's pension footnote for the value of excess pension assets.

¹⁰ Tax policy varies widely across countries. Check local tax policy to determine if, how, and when you can net past losses against future income.

¹¹ As detailed in Chapter 11, we classify deferred taxes into tax loss carryforwards, operating deferred taxes, and nonoperating deferred taxes. Only tax loss carryforwards should be valued separately. The other two accounts are either incorporated into free cash flow via cash operating taxes or valued as part of the account that generated the deferral.

available, year-by-year tax savings will be difficult to assess because tax loss carryforwards must be matched in the country in which they are generated. A pragmatic approach is to assume the tax benefits will be realized over an arbitrary period—say, five years. If your valuation of tax loss carryforwards affects share price in a meaningful way, ask management for additional disclosures regarding the location and timing of tax credits.

Finally, be careful not to double-count future tax savings by also incorporating them into the projected free cash flow. Since we value tax loss carryforwards separately, the tax loss carryforward is classified as a nonoperating asset and not included as part of either net operating profit after taxes (NOPAT) or invested capital.

VALUING INTEREST-BEARING DEBT

With enterprise value in hand, subtract the value of nonequity claims to determine equity value. Nonequity claims are found in the liability and equity sections of the balance sheet. Nonequity claims include traditional interest-bearing debt, debt equivalents such as unfunded retirement obligations, and hybrid securities that have characteristics of both debt and equity. In this section, we discuss traditional interest-bearing debt.

Traditional debt comes in many forms: commercial paper, notes payable, fixed and floating bank loans, corporate bonds, and capitalized leases. For companies with investment-grade debt, the value of debt will be independent of the value of operations. Consequently, each security's value can be estimated separately. For highly levered companies and companies in distress, this is not the case. In these situations, the value of debt will be linked to value of core operations, and both values must be determined concurrently.

Investment-Grade Debt If the debt is relatively secure and actively traded, use the market value of debt.¹² Market prices for U.S. corporate debt are reported on the Financial Industry Regulatory Authority (FINRA) Trade Reporting and Compliance Engine (TRACE) system.¹³ If the debt instrument is not traded, estimate current value by discounting the promised interest payments and the principal repayment at a yield to maturity that reflects the riskiness

¹² When a bond's yield is below its coupon rate, the bond will trade above its face value. Intuition dictates that, at most, the bond's face value should be deducted from enterprise value. Yet since enterprise value is computed using the cost of debt (via the weighted average of cost of capital) and not the coupon rate, subtracting face value is inconsistent with how enterprise value is computed. In cases where bonds are callable at face value, market prices will rarely exceed face value.

¹³ Developed by FINRA, the TRACE system facilitates the mandatory reporting of over-the-counter market transactions for eligible debt securities in the United States. It is available to the public via FINRA's website. At the time of publication, FINRA provided an online bond search tool on its home page.

of the debt—typically based on the company's bond rating. The book value of debt is a reasonable approximation for fixed-rate debt if interest rates and default risk have not significantly changed since the debt issuance. For floating-rate debt, value is not sensitive to interest rates, and book value is a reasonable approximation if the company's risk of default has been generally stable.

If you are using your valuation model to test changes in operating performance (for instance, a new initiative that will improve operating margins), the value of debt under your new assumptions may differ from its current market value. Always check leverage ratios, such as the interest coverage ratio, to test whether the company's bond rating will change under the new forecasts; often it will not. A change in bond rating can be translated into a new yield to maturity for debt, which in turn will allow you to revalue the debt. For more on debt ratings and interest rates, see Chapter 33.

Highly Levered Companies For companies with significant debt or companies in financial distress, valuing debt requires careful analysis. For distressed companies, the intrinsic value of the debt will be at a significant discount to its book value and will fluctuate with the value of the enterprise. Essentially, the debt has become like equity: its value will depend directly on your estimate for the enterprise value.

To value debt in these situations, apply an integrated-scenario approach. Exhibit 16.3 presents a simple two-scenario example for a company with significant debt. In scenario A, the company's management can implement improvements in operating margin, inventory turns, and so on. In scenario B, changes are unsuccessful, and performance remains at its current level.

For each scenario, estimate the enterprise value conditional on your financial forecasts.¹⁴ Next, deduct the *full value* of the debt and other nonequity claims from enterprise value. The full value is not the market value, but rather the value of debt if the company were default free.¹⁵ If the full value of debt is greater than enterprise value, set the equity value to zero. To complete the valuation, weight each scenario's resulting equity value by its probability of occurrence. For the company in Exhibit 16.3, scenario A leads to an equity valuation of \$300 million, whereas the equity value in scenario B is zero. If the probability of each scenario is 50 percent, the value of equity is \$150 million.

The scenario valuation approach treats equity like a call option on enterprise value. A more comprehensive model would estimate the entire distribution of potential enterprise values and use an option-pricing model, such as the Black-Scholes model, to value equity.¹⁶ Using an option-pricing model

¹⁴ All nonequity claims need to be included in the scenario approach for distressed companies. The order in which nonequity claims are paid upon liquidation will make a difference for the value of each nonequity claim but not for the equity value.

¹⁵ If the coupon rate does not equal the yield on comparable bonds, the value of debt will not equal the book value, even if the debt is default free.

¹⁶ Chapter 39 describes option-pricing models.

EXHIBIT 16.3 **Valuation of Equity Using Scenario Analysis**

\$ million					
	Enterprise value	Face value of debt	Equity value ¹	Probability of occurrence	Weighted equity value
Scenario A					
New owner successfully implements value improvements.	1,500	1,200	300	50%	150
Scenario B					
Company maintains current performance.	900	1,200	—	50%	—
				Equity value:	150

¹Equity value equals enterprise value less the face value of debt, or zero, whichever is greater.

rather than scenario analysis to value equity, however, has practical drawbacks. First, to model the distribution of enterprise values, you must forecast the expected change and volatility for each source of uncertainty, such as revenue growth and gross margin. This too easily becomes a mechanical exercise that replaces a thoughtful analysis of the underlying economics of potential scenarios. Second, most options models treat each source of uncertainty as independent of the others. This can lead to outcomes that are economically unrealistic. For these reasons, we believe a thoughtful scenario analysis will lead to a better-informed and more accurate valuation than an advanced options model will.

VALUING DEBT EQUIVALENTS

Debt equivalents have the characteristics of debt but are not formal loan contracts or traded securities. They include operating provisions such as plant decommissioning, nonoperating provisions such as restructuring charges, operating leases, and contingent liabilities such as pending lawsuits. We discuss the most common debt equivalents next.

Provisions

Certain provisions other than retirement-related liabilities must be deducted as debt equivalents. We distinguish four types of provisions (as introduced in Chapter 11 and discussed in detail in Chapter 21) and value them as follows:

1. Ongoing operating provisions (such as for warranties and product returns) are already accounted for in the free cash flows and therefore *should not be deducted* from enterprise value.

2. Long-term operating provisions (e.g., plant-decommissioning costs) *should be deducted* from enterprise value as debt equivalents. Because these provisions cover cash expenses that are payable in the long term, they are recorded at the discounted value in the balance sheet. In this case, there is no need to perform a separate DCF analysis, and you can *use the book value of the liability* in your valuation.¹⁷
3. Nonoperating provisions (in cases such as restructuring charges resulting from layoffs) *should be deducted* from enterprise value as a debt equivalent. Although a discounted value would be ideal, the book value from the balance sheet is often a reasonable approximation. These provisions are recorded on the financial statements at a nondiscounted value, because outlays are usually made in the near term.
4. Income-smoothing provisions should be eliminated from NOPAT. Consequently, they *should not be deducted* from enterprise value. For an example of income smoothing, see the sale-leaseback example for FedEx presented at the end of Chapter 11.

Leases

Starting in 2019, companies are required to recognize nearly all leases, including operating leases, on the balance sheet. For companies that report using IFRS, lease-related interest is recorded as a financial expense, and lease-related liabilities are incorporated within debt. Therefore, no adjustment is required.

For companies that report using U.S. GAAP, there are two types of leases: finance leases and operating leases. The treatment of finance leases is identical to IFRS, so no adjustment to enterprise value is required. In contrast, operating leases require special care. To determine equity value, remove embedded interest from operating expense, include the year-to-year change in “right-to-use” assets in free cash flow, and deduct the operating-lease liability from enterprise value to determine equity value.¹⁸ To value equity consistently, all three actions are required. If you choose not to adjust for embedded interest or include the change of “right-to-use” assets on free cash flow, do *not* subtract the value of operating leases.

Chapter 22 details the new accounting rules, required adjustments, and valuation of leases.

Unfunded Retirement Obligations

Unfunded retirement obligations, such as unfunded pensions and post-retirement medical benefits, should be treated as debt equivalents and

¹⁷ The company will also recognize a decommissioning asset at the time of initial investment. The decommissioning asset is already incorporated into free cash flow, so no adjustment for the asset is required.

¹⁸ For a more comprehensive summary, see the Operating Leases section of Chapter 11.

deducted from enterprise value to determine equity value. Since the future contributions to eliminate unfunded liabilities are tax deductible at the marginal tax rate, multiply unfunded pension liabilities by 1 minus the statutory income tax rate. For details on pension accounting and valuation, see Chapter 23.

Contingent Liabilities

Certain liabilities are not disclosed in the balance sheet but are separately discussed in the notes to the balance sheet. Examples are possible liabilities from pending litigation and loan guarantees. When possible, estimate each liability's expected after-tax cash flows (if the costs are tax deductible), and discount these at the cost of debt. Unfortunately, an external assessment of the probability of such cash flows materializing is challenging, so the valuation should be interpreted with caution. To provide some boundaries on your final valuation, estimate the value of contingent liabilities for a range of probabilities.

VALUING HYBRID SECURITIES AND NONCONTROLLING INTERESTS

For stable, profitable companies, the current values of debt and debt equivalents are typically independent of enterprise value. For hybrid securities and noncontrolling interests, this is not the case. Each must be valued in conjunction with estimates of enterprise value. The most common hybrid securities are convertible debt, convertible preferred stock, and employee stock options. We will detail the treatment of all three, as well as noncontrolling interests.

Convertible Securities

Convertible bonds are corporate bonds that can be exchanged for common equity at a predetermined conversion ratio.¹⁹ Convertible preferred stock has the same basic structure as convertible bonds, except convertible stock often comes with other rights of control, such as board seats. Both have become a major source of financing for publicly traded technology companies.²⁰ A convertible bond is essentially a package of a straight corporate bond plus a call option on equity (the conversion option). Because the conversion option can have significant value, this form of debt requires treatment different from that of regular corporate debt.

¹⁹ For more on convertible bonds, see R. Brealey, S. Myers, and F. Allen, *Principles of Corporate Finance*, 12th ed. (New York: McGraw-Hill, 2017), chap. 24.

²⁰ R. Molla, "Tech Companies Are Taking Out Record Amounts of Convertible Debt. Here's Why," *Vox*, June 20, 2018, www.vox.com.

The value of convertibles depends on the enterprise value. In contrast to valuation of straight debt, neither the book value nor the simple DCF value of bond cash flows is a good proxy for calculating the value of convertibles. Depending on the information available, there are four possible methods to apply:

1. *Fair value.* Companies report the “fair” value of financial instruments, including convertible debt, in the notes to the financial statements. Companies value these investments using quoted market prices or pricing models, and they disclose the methodology used. Use this value if enterprise value has not changed significantly since the last financial report.
2. *Market price.* Many convertible bonds are actively traded with quoted prices. For U.S. convertible debt, use the TRACE database to determine the market value of debt when the enterprise value has materially changed since the last filing.
3. *Black-Scholes value.* When the fair value or market value is inappropriate,²¹ we recommend using an option-based valuation for convertible debt. Accurate valuation of convertible bonds with option-based models is not straightforward. That said, by following methods outlined by DeSpiegeleer, Van Hulle, and Schoutens, you can make a reasonable approximation applying an adjusted Black-Scholes option-pricing model.²²
4. *Conversion value.* The conversion value approach assumes that all convertible bonds are immediately exchanged for equity and ignores the time value of the conversion option. It leads to reasonable results when the conversion option is deep in the money, meaning the bond is more valuable when converted into equity than when held for future coupon and principal payments.

Valuing Convertibles Exhibit 16.4 illustrates all four valuation methods for the mobile-payments company Square. Square has not issued traditional, fixed-payment debt. Instead, the company issued two convertible bonds: a \$211.7 million convertible bond due in March 2022 and an \$862.5 million convertible bond due in May 2023.²³ Because the coupon rate was below the prevailing yield for nonconvertible debt at the time of offering, the bonds are

²¹ If you plan to modify enterprise value because of proposed operating changes, the fair value is no longer appropriate, as the value of convertible debt will change with enterprise value.

²² For more on the valuation of convertible debt, see, for example, J. DeSpiegeleer, C. Van Hulle, and W. Schoutens, *The Handbook of Hybrid Securities: Convertible Bonds, CoCo Bonds, and Bail-In* (Hoboken, NJ: John Wiley & Sons, 2014).

²³ Square originally issued \$440 million in convertible bonds. Investors have exercised many of the 2022 convertible bonds, such that only \$211.7 million in principal remains as of year-end 2018.

reported on the balance sheet below their principal value, at \$181.2 million and \$718.5 million, respectively.²⁴

The first column in Exhibit 16.4 values Square's equity using the fair value of convertible debt reported in the company's 10-K. The second column presents the year-end closing price collected from the TRACE database. Compared with the book value reported on the balance sheet, the company's convertible debt trades at a significant premium. For instance, the convertible debt due in 2023 was valued by Square at \$901.5 million in December 2018 versus \$718.5 million in book value.

The significant premium to book value can be traced to the value of the conversion feature. According to Square's annual report, the bonds maturing in 2022 are convertible at \$22.95 per share.²⁵ At this conversion price, \$211.7 million in outstanding principal is convertible into 9.23 million shares. With Square's stock trading at \$56.09 in December 2018, the bonds can be converted into the equivalent of \$517.5 million in equity. The bond trades at a market price (\$523.2 million), which is slightly higher than the bond's conversion value (\$517.5 million), given the upside potential and downside protection the bond offers.

EXHIBIT 16.4 **Square Convertible Debt, December 2018**

\$ million						
Capital structure	Fair value ¹	Market price ²	Black-Scholes value ³	Conversion value	Carrying value	Principal outstanding
Enterprise value	26,300.0	26,300.0	26,300.0	26,300.0		
→ Convertible debt at 0.375% due 2022	(515.7)	(523.2)	(534.8)	—	181.2	211.7
→ Convertible debt at 0.5% due 2023	(901.5)	(899.2)	(917.9)	—	718.5	862.5
→ Convertible note hedge	230.9	230.9	230.9	—		
Employee options	(1,543.8)	(1,543.8)	(1,543.8)	(1,543.8)		
Equity value	23,570.0	23,564.8	23,534.5	24,756.2		
Number of shares, millions						
Number of nondiluted shares	419.7	419.7	419.7	419.7		
→ New shares issued	—	—	—	20.3		
Number of diluted shares	419.7	419.7	419.7	440.0		
Value per share, \$	56.1	56.1	56.0	56.3		

¹ Value of convertible bonds reported in 2018 10-K in note 5, "Fair Value of Financial Instruments," under "Fair Value (Level 2)."

² Market price reported by the FINRA TRACE database as of December 31, 2018.

³ Value estimated using Black-Scholes option-pricing model and company-disclosed inputs.

²⁴ When a company issues convertible debt at a coupon rate below the yield on similar nonconvertible debt, it will be recorded on the balance sheet at a discount but may not trade at a discount. This is because the conversion feature has value. The value of the conversion feature, however, is not recorded as part of debt, but rather as shareholders' equity. Since the book value of equity is not used in DCF valuation, this can lead to a significant underestimation of the convertible's value. For more on the accounting related to convertible debt, see Accounting Principles Board (APB) 14-1, "Accounting for Convertible Debt Instruments That May Be Settled in Cash upon Conversion (Including Partial Cash Settlement)." Financial Accounting Standards Board, May 9, 2008, www.fasb.org.

²⁵ Reported in Square's 2018 annual report, note 12, "Indebtedness."

If improvements to operations increase enterprise value, it becomes necessary to revalue Square's convertibles using an option-pricing model. To model the value of Square's convertible debt, disaggregate the value of convertible debt into underlying straight debt and the option value to convert. For the bond maturing in 2022, the value of straight debt equals the net present value of a 0.375 percent coupon bond yielding 2.48 percent (the yield on comparable bonds without conversion features), maturing in 3.25 years (the remaining life). Without conversion, this bond is valued at 93.45 percent of \$211.7 million in outstanding principal, or \$197.9 million.

To determine the option to convert's value, you need six inputs: the underlying asset value, the strike price, the volatility of the underlying asset, the risk-free rate, the time to maturity, and the dividend rate on the underlying asset. For the option embedded in Square's 2022 convertible bond, the underlying asset is 9.23 million shares of Square stock, whose current value equals \$517.5 million. The strike price, which represents what the investor must pay to receive the shares, equals the current value of straight debt, currently valued at \$197.9 million. The volatility of Square shares (30.9 percent) is reported in the company's 10-K. The bond's time to maturity is 3.25 years, and the current risk-free rate is 2.48 percent.²⁶ Square does not pay dividends, so the dividend yield is set at zero.

Plugging the data into a Black-Scholes estimator leads to an option value of \$336.9 million. Thus, as illustrated in the third data column of Exhibit 16.4, the Black-Scholes value of the convertible debt equals \$534.8 million (\$197.9 in straight debt plus \$336.9 in option value). This result is contingent on stability of the Black-Scholes inputs, especially volatility. If volatility is expected to drop as the company matures, the historical estimate of volatility will overestimate the option value. The errant valuation is largest for long-dated options, which is often the case for convertible debt.

An alternative to option pricing is the conversion value approach, shown in the fourth data column of Exhibit 16.4. The method is easier to implement than Black-Scholes but ignores optionality. Under the conversion value approach, convertible bonds are converted immediately into equity. Since Square's bonds are convertible into 20.3 million shares (9.2 million shares from the convertible debt due in 2022 and 11.1 million shares from the convertible debt due in 2023), non-diluted shares are increased from 419.7 million to 440.0 million. This approach zeroes out convertible debt and divides the equity value by diluted shares.

In this case, each approach leads to a similar value because the value of conversion is much higher than the value of traditional debt (known as being in the money). For bonds out of the money, the conversion approach will lead to an underestimation of the bonds' value. Therefore, we recommend using an option valuation model, such as Black-Scholes.

²⁶ Square's convertible debt is not callable, so the remaining maturity can be used in the options valuation. If the debt is callable, this must be incorporated into the bond's valuation.

Convertible Bond Hedges When a company issues a convertible bond, the bond is sometimes accompanied by a complex derivative transaction to effectively increase the strike price.²⁷ For example, Square, Etsy, and Twitter have all issued convertible debt with accompanying hedges. Investors prefer strike prices close to the current share price. Issuers, concerned about dilution from conversion into equity, prefer a higher strike price that lowers the odds of conversion.

In its annual report, Square reports, “The Company entered into convertible note hedge transactions . . . to effectively increase the overall conversion price from approximately \$22.95 per share to approximately \$31.18 per share.” To account for the value of the hedge, we use Black-Scholes to revalue the convertible bond at the higher strike price. The convertible note hedge reported in Exhibit 16.4 equals the difference between the original and synthetic bond price.

Although Square does not report the value of the hedge on the balance sheet or in the notes, the company does disclose that “the convertible note hedge and warrant transactions may affect the value of our Class A common stock.” Even with the recent improvements in accounting transparency, a diligent analysis of the notes continues to be critical!

Employee Stock Options

Many companies offer their employees stock options as part of their compensation. Options give the holder the right, but not the obligation, to buy company stock at a specified price, known as the exercise price. Since employee stock options have long maturities and the company’s stock price could eventually rise above the exercise price, options can have great value.

Employee stock options affect a company valuation in two ways. First, the value of options that will be *granted in the future* needs to be captured in the free-cash-flow projections or in a separate DCF valuation, following the guidelines in Chapter 11. If captured in the free-cash-flow projections, the value of future options grants is included in the value of operations and should not be treated as a nonequity claim. Second, the value of options *currently outstanding* must be subtracted from enterprise value as a nonequity claim. Note, however, that the value of the options will depend on your estimate of enterprise value. Your option valuation should reflect this.

The following approaches can be used for valuing employee options:

- *Company-disclosed fair value.* Start by searching the annual report for the company’s assessment of fair value. For instance, Square reports the “aggregate intrinsic value” of employee options at \$1.544 billion in the note on stockholders’ equity.

²⁷ In the transaction, the company purchases a call option on its own shares at the original share price and writes a second call option at the preferred conversion price.

- *Option-pricing model.* If the company's enterprise value has changed since the last financial filing, estimate the value using option valuation models such as Black-Scholes or more advanced binomial (lattice) models. Under U.S. GAAP and IFRS, the notes to the balance sheet report the total value of all employee stock options outstanding, as estimated by such option-pricing models. Note that the balance sheet value is a good approximation only if your estimate of share price is close to the one underlying the option values in the annual report. Otherwise, you need to create a new valuation using an option-pricing model.²⁸ The notes disclose the information required for valuation.
- *Exercise value approach.* The exercise value approach provides only a lower bound for the value of employee options, the smallest value that would round up to the estimated value. It assumes that all options are exercised immediately and thereby ignores the time value of the options. The resulting valuation error increases as options have longer time to maturity, the company's stock has higher volatility, and the company's share price is closer to the exercise price. Given that a more accurate valuation is already disclosed in the annual report, we do not recommend this method. However, it is still quite common among practitioners.

Exhibit 16.5 provides an example of the three valuation methods. The first data column is based on the fair value reported by the company, which it calls "aggregate intrinsic value." The second and third data columns use the Black-Scholes option-pricing model to value first the outstanding options and second the options that can be currently exercised. The value of outstanding options will be less than that of options that can be exercised, because outstanding options include some options that will be lost if the employee leaves the company.

To estimate the value of employee stock options, you need six inputs: the current stock price, the average strike price, the stock's volatility, the risk-free rate, the time to maturity, and the stock's dividend rate. Square's current share price equals \$56.09. The other inputs are disclosed in Square's 10-K for both outstanding and exercisable options. For outstanding options, the weighted average strike price equals \$9.52, the volatility of Square's shares equals 30.9 percent, and the average time to maturity is reported at 5.45 years. The current risk-free rate over five years is 2.51 percent, and the expected dividend rate is zero. The Black-Scholes estimator prices the average option at \$47.81.²⁹

²⁸ For more on the valuation of employee stock options, see, for example, J. Hull and A. White, "How to Value Employee Stock Options," *Financial Analysts Journal* 60, no. 1 (January/February 2004): 114–119.

²⁹ Using Black-Scholes to determine the value of a single option on an average strike price will undervalue a portfolio of options with a spread of strike prices. Unless you know the spread of strike prices, you cannot measure the bias.

EXHIBIT 16.5 Square Employee Options, December 2018

\$ million

Capital structure	Aggregate intrinsic value ¹	Using Black-Scholes ²		Exercise value approach
		Value of outstanding options	Value of exercisable options	
Enterprise value	26,300.0	26,300.0	26,300.0	26,300.0
Convertible debt at 0.375% due 2022	(515.7)	(515.7)	(515.7)	(515.7)
Convertible debt at 0.5% due 2023	(901.5)	(901.5)	(901.5)	(901.5)
Convertible note hedge	230.9	230.9	230.9	230.9
→ Employee options: value	(1,543.8)	(1,584.9)	(1,507.9)	—
→ Employee options: exercise proceeds	—	—	—	315.6
Equity value	23,570.0	23,528.9	23,605.8	25,429.4
Number of shares, millions				
Number of nondiluted shares	419.7	419.7	419.7	419.7
→ New shares issued	—	—	—	33.2
Number of diluted shares	419.7	419.7	419.7	452.8
Value per share, \$	56.1	56.0	56.2	56.1

¹ Value of options reported in 2018 10-K in note 15, "Shareholder's Equity," under "Aggregate Intrinsic Value."² Value estimated using Black-Scholes option-pricing model and company-disclosed inputs.

With 33.2 million options outstanding, the aggregate value of options is valued at \$1.58 billion. To estimate share price, deduct the aggregate value from enterprise value, and divide by the number of undiluted shares. Since some outstanding options will go unclaimed, repeat the process for just the options that can be exercised. The actual value will fall somewhere between the two.

Under the exercise value approach, employee options are assumed to be exercised immediately. According to Square's 2018 10-K, 33,152,881 shares can be exercised at an average strike price of \$9.52, for total proceeds of \$315.6 million. Exercise of employee options generates cash for the company and increases shares outstanding from 419.7 million to 452.8 million. Dividing equity value by diluted shares leads to a value of \$56.1, slightly lower than the value under the Black-Scholes method.

Noncontrolling Interests by Other Companies

When a company controls a subsidiary but does not fully own it, the subsidiary's financial statements must be fully consolidated in the group accounts. The subsidiary's assets and liabilities will be indistinguishable from the parent company's accounts, but the portion of the subsidiary's equity not owned by the parent company will be separated from other equity accounts as noncontrolling interest.³⁰ Since the full value of the subsidiary will be incorporated

³⁰ For example, Berkshire Hathaway reported \$3.8 billion in noncontrolling interests in 2018. This amount can be found on the company's balance sheet under shareholders' equity.

into the value of operations, a valuation adjustment must be made for the portion of the subsidiary not owned by the parent company being valued.

Because noncontrolling interests by other companies are to a certain extent the mirror image of nonconsolidated assets, the recommended valuation approach for noncontrolling interests is similar to that of nonconsolidated assets, described earlier in this chapter. In the case of a minority carve-out (in which the consolidated but not fully owned subsidiary is publicly traded), deduct the proportional market value owned by outsiders from enterprise value to determine equity value. Alternatively, you can perform a separate valuation using a DCF approach, multiples, or a tracking portfolio, depending on the amount of information available. Remember, however, that a noncontrolling interest is a claim on a subsidiary, not the entire company. Thus, any valuation should be directly related to the subsidiary and not to the company as a whole.

ESTIMATING VALUE PER SHARE

The final step in a valuation is to calculate the value per share. Assuming that you have used an option-based valuation approach for convertible bonds and employee options, divide the total equity value by the number of *undiluted* shares outstanding. Use the undiluted (rather than diluted) number of shares because the full values of convertible debt and stock options have already been deducted from the enterprise value as nonequity claims. Also, use the most recent number of undiluted shares outstanding. Do not use the weighted average of shares outstanding; it is reported in the financial statements to determine average earnings per share.

The number of shares outstanding is the gross number of shares issued, less the number of shares held in treasury. Most U.S. and European companies report the number of shares issued and those held in treasury under shareholders' equity. However, some companies report treasury shares as an investment asset, which is incorrect from an economic perspective. Treat them instead as a reduction in the number of shares outstanding.

If you used the conversion and exercise value approach to account for employee options and convertible debt and stock options, divide by the diluted number of shares.

With intrinsic value per share in hand, you have completed the mechanics of your valuation. But the job is not done. The next two chapters discuss how to stress-test your valuation using integrated scenarios and trading multiples.

REVIEW QUESTIONS

1. PhoneCo, a U.S. maker of smartphones and other consumer products, has \$60 billion of cash and marketable securities on its balance sheet and revenues of \$50 billion. Using the rule of thumb mentioned in the text, how much would you estimate as cash needed for operations and excess cash and marketable securities? Now suppose that you discover that of the \$60 billion in cash, \$40 billion of it is from profits earned overseas. Would your estimate of the value of excess cash and marketable securities change? How does this affect your valuation of PhoneCo?
2. MarineCo manufactures, markets, and distributes recreational motorboats. Using discounted-free-cash-flow analysis, you value the company's operations at \$2,500 million. The company has a 20 percent stake in a nonconsolidated subsidiary valued at \$500 million. The investment is recorded on MarineCo's balance sheet as an equity investment of \$50 million. MarineCo is looking to increase its ownership. The company's marginal tax rate is 30 percent. Based on this information, what is MarineCo's enterprise value? If new management announced its plan to sell the company's stake in the subsidiary at its current value, how would that change your valuation?
3. MarineCo has no debt but does have unfunded pension liabilities valued at \$200 million, recorded as a long-term other liability. MarineCo has detailed in its annual report a potential legal judgment of \$100 million for defective engines. Since management estimates a 90 percent likelihood that the judgment will be enforced against the engine maker and not MarineCo, it has not reported a liability on the balance sheet. The company's marginal tax rate is 30 percent. Based on this information and information provided in Question 2 (assuming no sale of the subsidiary), what is MarineCo's equity value?
4. You are valuing a company that has \$200 million in debt using probability-weighted-scenario analysis. You carefully model three scenarios, such that the resulting enterprise value equals \$300 million in scenario 1, \$200 million in scenario 2, and \$100 million in scenario 3. The probabilities of the scenarios are 25 percent, 50 percent, and 25 percent, respectively. What is the expected enterprise value? What is the expected equity value? Management announces a new plan that eliminates the downside scenario (scenario 3), yielding a 75 percent probability of scenario 2. What happens to enterprise value and equity value? Why does enterprise value rise more than equity value?
5. You are valuing a technology company whose enterprise value is \$800 million. The company has no debt but considerable employee options, 10 million in total. Based on option-pricing models, you value each option at \$6.67 per option. Assume that the average strike price equals \$15. If the company has 40 million shares outstanding, what is the

company's equity value and value per share using (a) the option-pricing model and (b) the exercise value approach? Why is the option-pricing model the preferred method?

6. You are valuing an industrial firm whose enterprise value is \$10 billion. The company has no straight debt but does have 100,000 convertible bonds outstanding. The market value of each bond is \$1,150. If the company has 500 million shares outstanding, what is the company's equity value and value per share using (a) the market value approach and (b) the conversion value approach?

